Vedant Bhadri

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EDUCATION

Auckland University of Technology, Master of Computer and Information Sciences *July 2025 - Present*
Relevant Coursework: Artificial Intelligence, Deep Learning, Data Mining and Machine Learning, Neuroinformatics, Nature Inspired Computing
Member: AUT Computer Science and Engineering Association (CSEA), AUT Off-Grid Club

KLS Gogte Institute of Technology, Bachelor of Engineering in Information Science and Engineering *Nov 2020 – July 2024*
Relevant Coursework: Data Structures and Algorithm, Data Science, Artificial Intelligence, Python Programming, Soft Computing, Robotic Process Automation, Database Management, Computer Networks, Operating Systems.
Member: Rotaract Club of GIT, GIT Association of Computer Machinery

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS, Bash
AI and Machine Learning: PyTorch, TensorFlow, scikit-learn, FAISS, NeuCom/NeuCube
Technologies: AWS (Bedrock, Textract), OpenAI API, REST APIs, Linux, Git, Google Colab, Raspberry Pi, Postman, VS Code

WORK EXPERIENCES

Auckland University of Technology **Auckland, New Zealand**
Teaching Assistant *Feb 2026 - Present*
Course: COMP501 Computing Technology in Society | COMP517 Data Analysis

- Support undergraduate students in weekly lab sessions, helping them work through experiments, data analysis, and problem-solving.
- Grade assignments and lab reports, giving clear, constructive feedback that helps students improve.
- Work with students across a wide range of technical skill levels, which has sharpened both my communication and adaptability.

Wynby Technologies **Bangalore, India**
Software Engineering Intern | Python, React APIs, SQL *Jun 2024 – Sept 2024*

- Built backend infrastructure in Python for a live algorithmic trading system, including REST APIs and SQL-based data pipelines to handle real-time market data.
- Designed trading logic, including trailing stop-loss mechanisms and pattern-recognition algorithms, to support automated strategy execution and keep downside risk in check.
- Worked closely with traders and engineers to get backend functionality into production, helping keep the system reliable under live, fast-moving market conditions.

PROJECTS

AI Powered Allergen Compliance & Menu Translation | AWS AI Innovation Mentorship | 3rd Place *Sept 2026*
AWS, Amazon Bedrock & Flask

- Worked in a team to build a web application that converts uploaded restaurant menus into multilingual digital menus with allergen information based on New Zealand food-labelling standards.
- Implemented the menu upload and OCR workflow using Amazon Textract, including the parsing logic that separates dish names and descriptions while filtering out prices, headings, and unrelated text.
- Added WebP image support, faster parallel dish processing, automated parser tests, and the option to clear old menu data before processing a new upload.

Non-Invasive Blood Glucose Estimation from PPG | Master's Thesis | Ongoing *Feb 2026*
Python, PyTorch, Signal Processing, Google Colab, Git, Raspberry Pi.

- Investigated the feasibility of estimating blood glucose non-invasively using wrist-based PPG/BVP signals collected from wearable devices, with a focus on developing a practical approach for continuous monitoring.
- Developed an end-to-end deep-learning pipeline covering signal-quality assessment, physiological feature extraction and a hybrid CNN–Transformer model. Evaluated its ability to generalise to unseen participants using subject-independent testing.
- Improved the project through systematic data-quality analysis, multiple modelling experiments and clinically relevant evaluation methods. Also prepared the trained model for real-time testing with an Empatica wearable and deployment on a Raspberry Pi.

Mini AI Coding Agent | Personal Project *Sept 2026*

Python, OpenAI API

- Built a terminal-based coding assistant that can inspect files, edit text files, and run shell commands to help complete user requests.
- Implemented an agent loop using OpenAI function calling: the model selects a tool, the Python application executes it, and the result is returned to the model for the next step.
- Added session conversation history, command confirmation, and environment-based API key loading to make the agent more practical to use.

EEG-Based ADHD Classification: ML & Spiking Neural Networks | Neuroinformatics | AUT *Sept 2025*

Python, Scikit-Learn, Neucom, Neucube,

- Classified ADHD vs. control subjects using 19-channel EEG theta-band features.
- Tried several models such as SVM, MLP, an Evolving Classification Function then went further and built a Spiking Neural Network, encoding the EEG into spike trains and mapping it onto a 3D brain model.
- Tested each approach on hold-out data to see which one generalised best, rather than assuming the fancier model would automatically win.

Smart-City Delivery Route Optimisation Using Genetic Algorithms | Nature Inspired Computing | AUT *Oct 2025*

Python, NumPy

- Worked on the Capacitated Vehicle Routing Problem with Time Windows, a classic hard logistics problem, under realistic smart-city constraints.
- Built an Enhanced Genetic Algorithm with adaptive mutation and 2-Opt local search to get better routes than a standard GA would find.
- Compared it directly against a baseline GA to check the added mechanisms were making routes better, not just taking longer to run.

Semantic Image Retrieval Using CNN Embeddings & FAISS | Artificial Intelligence | AUT *Oct 2025*

Python, TensorFlow, FAISS

- Built a system that finds visually similar images in a large dataset, given a query image.
- Used a pretrained ResNet-50 to pull out feature embeddings, and FAISS to search through them quickly at scale.
- Compared a few different similarity metrics (cosine, Euclidean, colour histogram) head-to-head before settling on the one that retrieved the best results.

AWARDS AND ACHIEVEMENTS

AWS AI Innovation Mentorship Programme (2026) | 3rd Place

Recognised for developing an AI-Powered menu processing and translation application using AWS services as a part of a team.

AUT International Scholarships – South Asia

Awarded in recognition of academic merit for postgraduate study at Auckland University of Technology

LEADERSHIP AND VOLUNTEERING EXPERIENCE

Off-Grid Club | Auckland University of Technology **Auckland, NZ**

General Executive *Apr 2026 – Present*

- Co-organised hiking, camping, and outdoor adventure events as part of the club's executive team, from planning logistics through to running the day itself.
- Managed participant registrations and grouped students into teams, keeping events running smoothly for everyone involved.

New Zealand Red Cross **Auckland, NZ**

Volunteer Retail Assistant *Dec 2025 –Feb 2026*

- Operated the checkout and handled cash and EFTPOS transactions accurately, providing friendly, reliable service to customers.
- Managed donated stock and maintained shop presentation, working closely with the team to keep the store running smoothly.

Rotaract Club of GIT **Belgaum, IN**

Volunteer *Jan 2021 – July 2024*

- Took part in community service projects aimed at creating positive, lasting impact in the local community.
- Supported fundraising initiatives, contributing to the club's efforts to resource and sustain its community work.